DESIGN FOR MANUFACTURABILITY (DFM) REPORT

Part: test_box | **Evaluation Date:** 2026-08-16 20:24

Material: ABS	**Units:** mm	**Surface Texture:** Plain	**Draft Threshold:** ≥ 1.0°

1. Part Summary & Dimensions

Total Volume:	1000.00 mm³	**Bounding Box (X × Y × Z):**	10.00 × 10.00 × 10.00 mm
Total Surface Area:	600.00 mm²	**Topology Counts:**	6 Faces, 12 Edges

2. Mold Pull Direction & 6-Axis Evaluation

Recommended Mold Opening Vector: **(1.00, 0.00, 0.00)** | **Algorithm:** 6-axis-cardinal-score | **Confidence:** HIGH

Candidate Axis	Undercut Faces	Zero Draft Faces	Recommendation
(+1, +0, +0)	1	4	**RECOMMENDED**
(-1, +0, +0)	1	4	Alternative
(+0, +1, +0)	1	4	Alternative
(+0, -1, +0)	1	4	Alternative
(+0, +0, +1)	1	4	Alternative
(+0, +0, -1)	1	4	Alternative

3. Draft Angle Analysis

Threshold Target: ≥ 1.0° (Plain finish) | **Pass:** **1** | **Low Draft:** **0** | **Zero Draft:** **4** | **Negative Draft:** **1** (Total: 6 faces)

Face ID	Draft Angle (°)	Normal Vector (Nx, Ny, Nz)	Draft Status
0	+90.00°	(1.00, -0.00, -0.00)	PASS (≥ 1.0°)
1	**-90.00°**	(-1.00, 0.00, 0.00)	**NEGATIVE_DRAFT (BLOCKS EJECTION)**
2	**+0.00°**	(0.00, 1.00, -0.00)	**ZERO_DRAFT (VERTICAL WALL)**
3	**-0.00°**	(-0.00, -1.00, 0.00)	**ZERO_DRAFT (VERTICAL WALL)**
4	**-0.00°**	(-0.00, -0.00, -1.00)	**ZERO_DRAFT (VERTICAL WALL)**
5	**+0.00°**	(0.00, 0.00, 1.00)	**ZERO_DRAFT (VERTICAL WALL)**

4. Undercuts & Resolution Plan

No undercut conditions detected. All part geometry will freely demold without side-actions, lifters, or complex slide tooling.

5. Parting Line & Shut-Off Quality

Loop Closure Status:	**CLOSED LOOP (Valid)**	**Planarity & Shut-off:**	Planar 2D parting line
Flashing Risk:	**LOW**	**Loop Vertex Count:**	5 points

Tooling Note: Planar parting line with uniform shut-off and minimal flash risk.

6. Mold Assembly & Tooling Split Volumes

Tooling Component	Volume (mm³)	Volume % of Stock Block	Ejection / Action Role
Master Stock Block	8000.00	100.0 %	Raw Tooling Steel/Aluminum Billet Envelope
Top Cavity Block (A-Side)	98.17	1.2 %	Forms cosmetic class-A outer surface
Bottom Core Block (B-Side)	3200.00	40.0 %	Forms functional interior & houses ejector system
Side Slider (Left / -X)	1850.91	23.1 %	Lateral mechanical slide for undercut extraction
Side Slider (Right / +X)	1850.91	23.1 %	Lateral mechanical slide for undercut extraction

7. Extended DFM Analysis (Physics & Moldability)

7.1 Wall Thickness & Sink Mark Evaluation

Nominal Thickness: 10.00 mm | **Min:** 10.00 mm | **Max:** 10.00 mm | **Variation:** **0.0%** (PASS) | *(Raycast Heuristic • Confidence: MEDIUM)*

7.2 Injection Gate Placement Candidates

Candidate	Gate Type	Location (X, Y, Z)	Suitability	Design Rationale
Gate #1 (Face 0)	Pin-point / Direct Sprue Gate	(5.0, -0.0, 5.0)	**91 / 100**	Accessible planar exterior surface (Pin-point / Direct Sprue Gate) with direct mold flow path into nominal wall.
Gate #2 (Face 2)	Edge / Submarine Gate at Parting Line	(0.0, 5.0, 5.0)	**86 / 100**	Accessible planar exterior surface (Edge / Submarine Gate at Parting Line) with direct mold flow path into nominal wall.
Gate #3 (Face 3)	Edge / Submarine Gate at Parting Line	(-0.0, -5.0, 5.0)	**86 / 100**	Accessible planar exterior surface (Edge / Submarine Gate at Parting Line) with direct mold flow path into nominal wall.

7.3 Rib/Boss Proportions, Corner Radii & Demolding

Rib / Boss Sizing:	No distinct cylindrical boss or isolated rib features identified on this geometry.
Corner Radii / Stress:	Min measured fillet radius: **0.00mm** (Target: ≥ 5.00mm). Internal radii should be >= 5.0 mm (0.5x nominal wall) to mitigate notch stress and resin flow restriction.
Ejector Pin Layout:	Identified **4** stable core-side candidate locations away from thin walls.

8. Known Gaps & Explicit Limitations

In accordance with rigorous engineering practice, any physical phenomena uncomputable from static BRep CAD topology are transparently documented below rather than approximated:

Analysis Scope / Domain	Technical Reason & Derivation Limitation
Volumetric Warpage & Material Shrinkage PVT Compensation	Uniform nominal geometry analyzed without non-isotropic polymer PVT shrinkage curves for material 'ABS'. Requires nonlinear finite-element injection molding warpage solver.
Cooling Circuit Conformal Optimization & Cycle Time Thermal Analysis	MHD/thermal transient CFD cooling simulation kernel is not integrated; cooling line optimization requires 3D transient heat flux modeling.
Weld Line Formation & Air Trap Location Prediction	Requires dynamic multi-phase cavity filling flow front simulation (Moldflow/OpenFOAM), uncomputable from static BRep topology alone.
Freeform Internal Rib Segmentation	Parametric CAD feature tree unavailable in STEP neutral format; ribs without distinct planar-pair boundary representations could not be isolated.